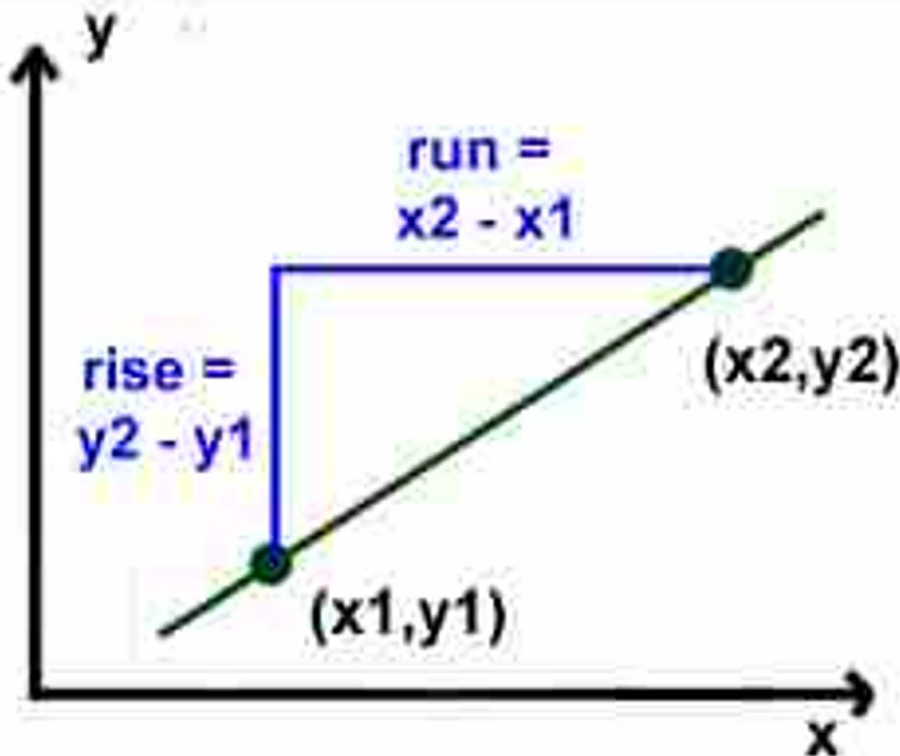

ecture 05 Differentiation/ Integration, Differential Equations, and Symbolic Math

3/4/ 2022

Outline

- (1) Linear Algebra
- (2) Polynomials
- (3) Optimization
- (4) Differentiation/Integration**
- (5) Differential Equations

Differentiation



$$\text{Slope} = (y_2 - y_1) / (x_2 - x_1)$$

Numerical Differentiation

- MATLAB can 'differentiate' numerically

- » `x=0:0.01:2*pi;`

- » `y=sin(x);`

- » `dydx=diff(y)./diff(x);`

- `diff` computes the first difference

- Can also operate on matrices

- » `mat=[1 3 5;4 8 6];`

- » `dm=diff(mat,1,2)`

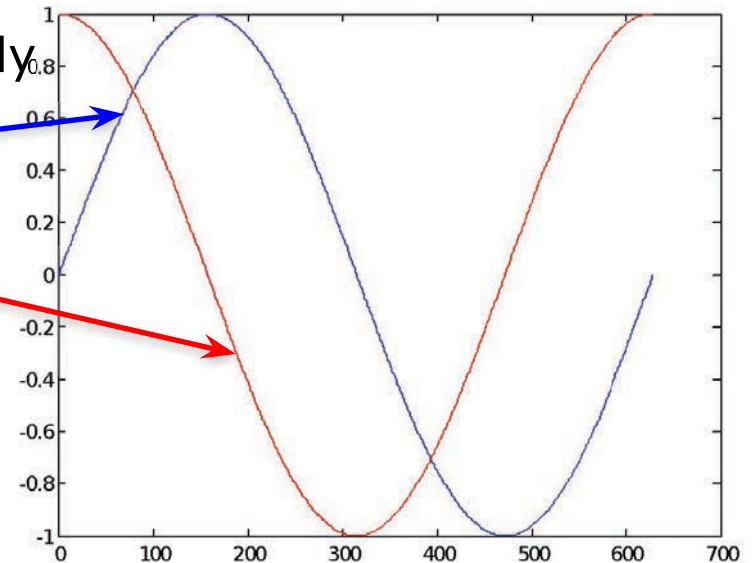
- first difference of `mat` along the 2nd dimension, `dm=[2 2;4 -2]`

- The opposite of `diff` is the cumulative sum `cumsum`

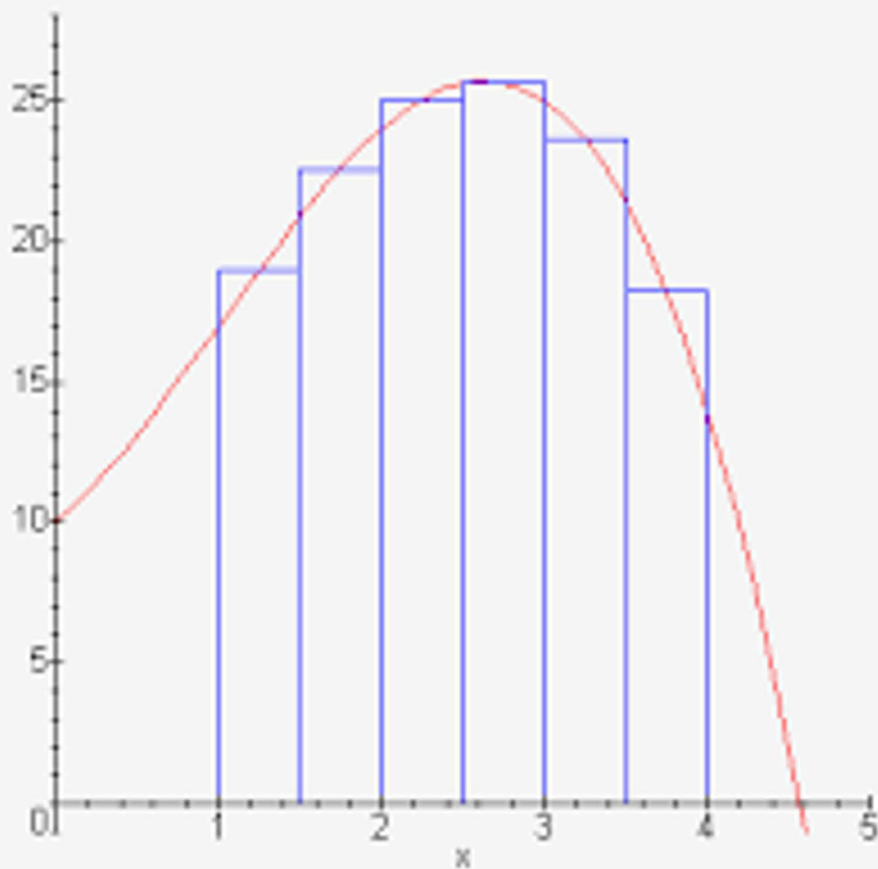
- 2D gradient

- » `[dx,dy]=gradient(mat);`

- Higher derivatives / complicated problems: Fit spline (see **help**)



Integration



Numerical Integration

- MATLAB contains common integration methods
- Adaptive Simpson's quadrature (input is a function)
 - » `q=quad('myFun',0,10)`
 - q is the integral of the function `myFun` from 0 to 10
 - » `q2=quad(@(x) sin(x).*x,0,pi)`
 - q2 is the integral of `sin(x).*x` from 0 to pi
- Trapezoidal rule (input is a vector)
 - » `x=0:0.01:pi;`
 - » `z=trapz(x,sin(x))`
 - z is the integral of `sin(x)` from 0 to pi
 - » `z2=trapz(x,sqrt(exp(x))./x)`
 - z2 is the integral of $\sqrt{e^x}/x$ from 0 to pi

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Differential Equations

Solve the first-order differential equation $\frac{dy}{dt} = ay$ with the initial condition $y(0) = 5$.

Specify the initial condition as the second input to `dsolve` by using the `==` operator. Specifying condition eliminates arbitrary constants, such as `C1`, `C2`, ..., from the solution.

```
syms y(t) a
eqn = diff(y,t) == a*y;
cond = y(0) == 5;
ySol(t) = dsolve(eqn,cond)
```

$$ySol(t) = 5 e^{at}$$

Next, solve the second-order differential equation $\frac{d^2y}{dt^2} = a^2y$ with the initial conditions $y(0) = b$ and $y'(0) = 1$.

Specify the second initial condition by assigning `diff(y,t)` to `Dy` and then using `Dy(0) == 1`.

```
syms y(t) a b
eqn = diff(y,t,2) == a^2*y;
Dy = diff(y,t);
cond = [y(0)==b, Dy(0)==1];
ySol(t) = dsolve(eqn,cond)
```

$$ySol(t) = \frac{e^{at} (ab + 1)}{2a} + \frac{e^{-at} (ab - 1)}{2a}$$

Outline

(1) Symbolic Math

(2) Simulink

(3) Data Structures

(4) File I / O

Symbolic Math Toolbox

- Symbolics vs. Numerics

	Advantages	Disadvantages
Symbolic	• Analytical solutions • Lets you intuit things about solution form	• Sometimes can't be solved • Can be overly complicated
Numeric	• Always get a solution • Can make solutions accurate • Easy to code	• Hard to extract a deeper understanding • Num. methods sometimes fail • Can take a while to compute

Symbolic Variables

- Symbolic variables are a type, like **double** or **char**
- To make symbolic variables, use **sym**
 - » **a=sym('1/3');**
 - » **b=sym('4/5');**
 - » **mat=sym([1 2;3 4]);**
 - fractions remain as fractions
 - » **c=sym('c','positive');**
 - can add tags to narrow down scope
 - see **help sym** for a list of tags
- Or use **syms**
 - » **syms x y real**
 - shorthand for **x=sym('x','real');** **y=sym('y','real');**

Symbolic Expressions

- Multiply, add, divide expressions

» **d=a*b** 

d =
4/15

☆ does $1/3 * 4/5 = 4/15$;

» **expand((a-c)^2);**

➤ multiplies out 

ans =
 $1/9 - 2/3 * c + c^2$

» **factor(ans)** 

ans =
 $1/9 * (3 * c - 1)^2$

➤ factors the expression

» **matInv=inv(mat)** 

ans =
[-2, 1]
[3/2, -1/2]

➤ Computes inverse symbolically

Cleaning up Symbolic Statements

» **pretty(ans)**

➤ makes it look nicer



$\frac{1}{9} - \frac{2}{3}c + c^2$

» **collect(3*x+4*y-1/3*x^2-x+3/2*y)**

➤ collects terms



$\text{ans} =$ $2x + \frac{11}{2}y - \frac{1}{3}x^2$
--

» **simplify(cos(x)^2+sin(x)^2)**

➤ simplifies expressions



$\text{ans} =$ 1

» **subs('c^2',c,5)**

➤ Replaces variables with numbers



$\text{ans} =$ 25

or expressions. To do multiple substitutions

pass a cell of variable names followed by a cell of values

» **subs('c^2',c,x/7)**



$\text{ans} =$ $\frac{1}{49}x^2$

More Symbolic Operations

- We can do symbolics with matrices too

» `mat=sym(' [a b;c d]');`

» `mat2=mat*[1 3;4 -2];`

➤ compute the product

```
mat2 =  
[      a+4*b,  3*a-2*b]  
[      c+4*d,  3*c-2*d]
```

» `d=det(mat)`

➤ compute the determinant

```
d =  
a*d-b*c
```

» `i=inv(mat)`

➤ find the inverse

```
i =  
[  d/(a*d-b*c), -b/(a*d-b*c)]  
[ -c/(a*d-b*c),  a/(a*d-b*c)]
```

- You can access symbolic matrix elements as before

» `i(1,2)`

```
ans =  
-b/(a*d-b*c)
```

Exercise: Symbolics

- The equation of a circle of radius r centered at (a,b) is given by: $(x-a)^2 + (y-b)^2 = r^2$
- Use **solve** to solve this equation for x and then for y
- It's always annoying to integrate by parts. Use **int** to do the following integral symbolically and then compute the value by **subs**tituting 0 for a and 2 for b :
$$\int_a^b x e^x dx$$

Exercise: Symbolics

- The equation of a circle of radius r centered at (a,b) is given by: $(x-a)^2 + (y-b)^2 = r^2$
- Use **solve** to solve this equation for x and then for y

» **syms a b r x y**

» **solve('(x-a)^2+(y-b)^2=r^2','x')**

» **solve('(x-a)^2+(y-b)^2=r^2','y')**

- It's always annoying to integrate by parts. Use **int** to do the following integral symbolically and then compute the value by **substituting** 0 for a and 2 for b :

$$\int_a^b x e^x dx$$

» **Q=int(x*exp(x),a,b)**

» **ans1 =subs(Q,{a,b},{0,2})**

» **ans2 = double(ans1)**